



LS-CAT

Life Sciences Collaborative Access Team

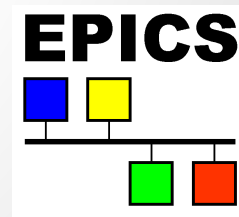


MXCuBE Status Report

Joseph Brunzelle



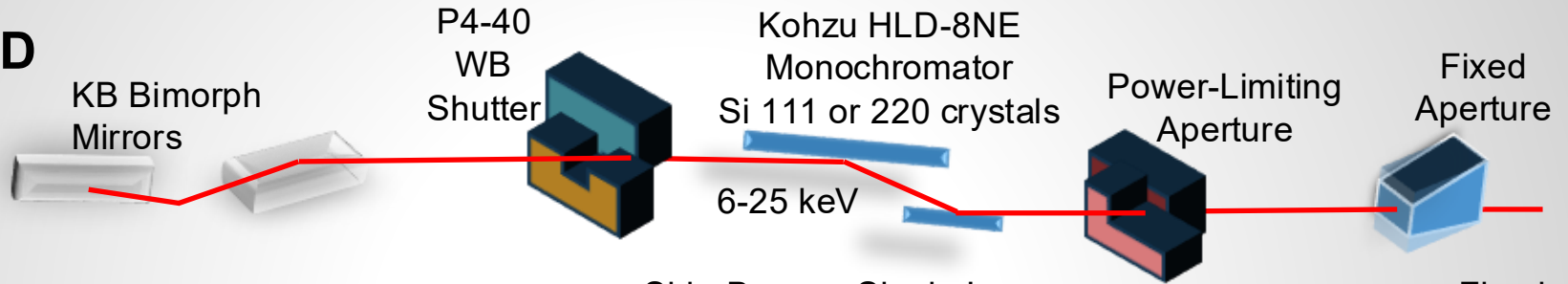
Argonne  NATIONAL LABORATORY | Advanced
Photon Source



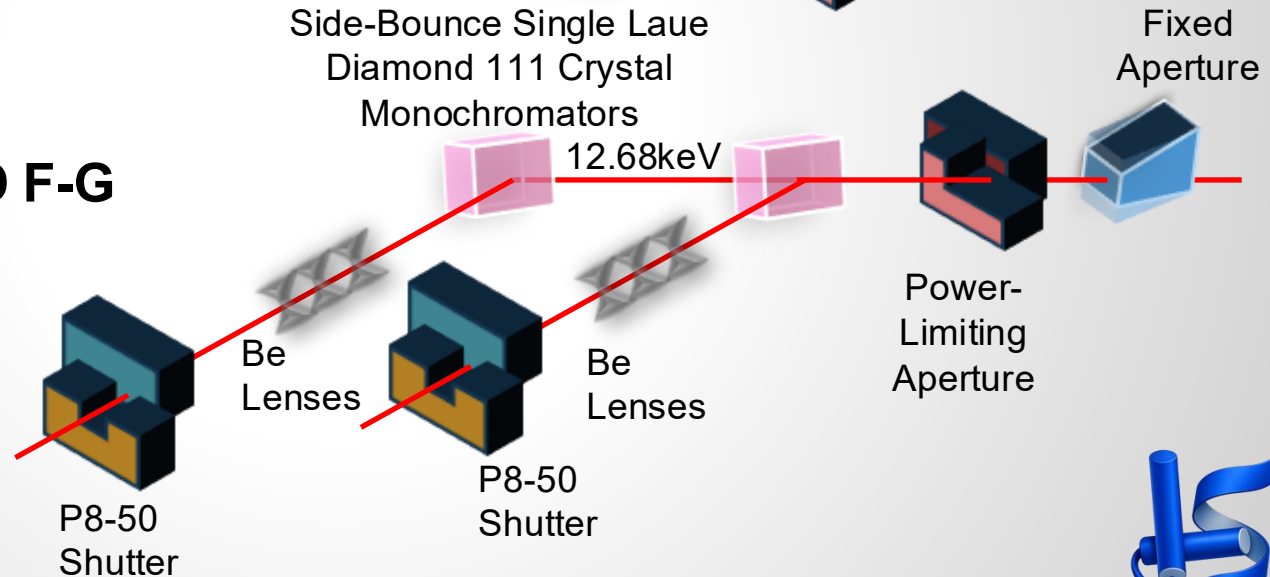
MxCuBE Pre-Meeting May 21, 2026

Beamline Layouts

21ID-D



21ID F-G



Clean Slate

- Removed MD2
- Removed CATS Robot System



MxCuBE Pre-Meeting May 21, 2026



End Station Assembly

- Installed MD3UP diffractometer
- Installed EIGER2 X 16M detector
- Installed UR7 robot + 37-puck dewar (592 samples)



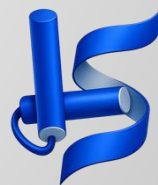
MXCuBE3 on 21-ID-D: Current Status

- **Software Stack**

- MXCuBEweb 4.622.0 | MXCuBEcore 1.444.0 | Python 3.11
- Interfaces: Exporter (MD3UP), EPICS (beamline PVs & UR7 robot), SIMPLON (EIGER2)

- **Working Capabilities**

- Sample mounting/dismounting via UR7 robot
- Visual sample centering
- Characterization collections (4-wedge orthogonal)
- Standard OSC data collections (exts trigger mode)
- ISPyB/SynchWeb metadata logging



MXCuBE Views

MXCuBE-Web (OSC) Samples Data collection Equipment ? Help Remote Session (cm00001) Sign out (b86669)

Beamline Cameras ▾ **Energy:** 11.9991 keV **Resolution:** 5.062 Å **Transmission:** 100.0 % **Cryo:** 200.00 k Sample Changer Detector Capillary Fast Shutter Safety shutter Ring Current

Beamline Actions ▾ **Wavelength:** 1.03 Å **Detector:** 750.0 mm **Intensity:** 2.57e+2 I/C **READY** **READY** **UNKNOWN** **CLOSED** **CLOSED** **0.01 mA**

Phase Control
Transfer ▾

Beam size
A50 ▾

Omega
360.00 ▾ 90 ▾

Y
0.096 ▾ 0.1 mm

Z
-0.630 ▾ 0.1 mm

Sample alignment
⬆️
⬅️ ⚙️ ➡️
⬇️
⚙️ Show motors ▾

Snapshot Draw grid 3-click centring Focus Zoom (LEVEL1) Backlight Frontlight Video size

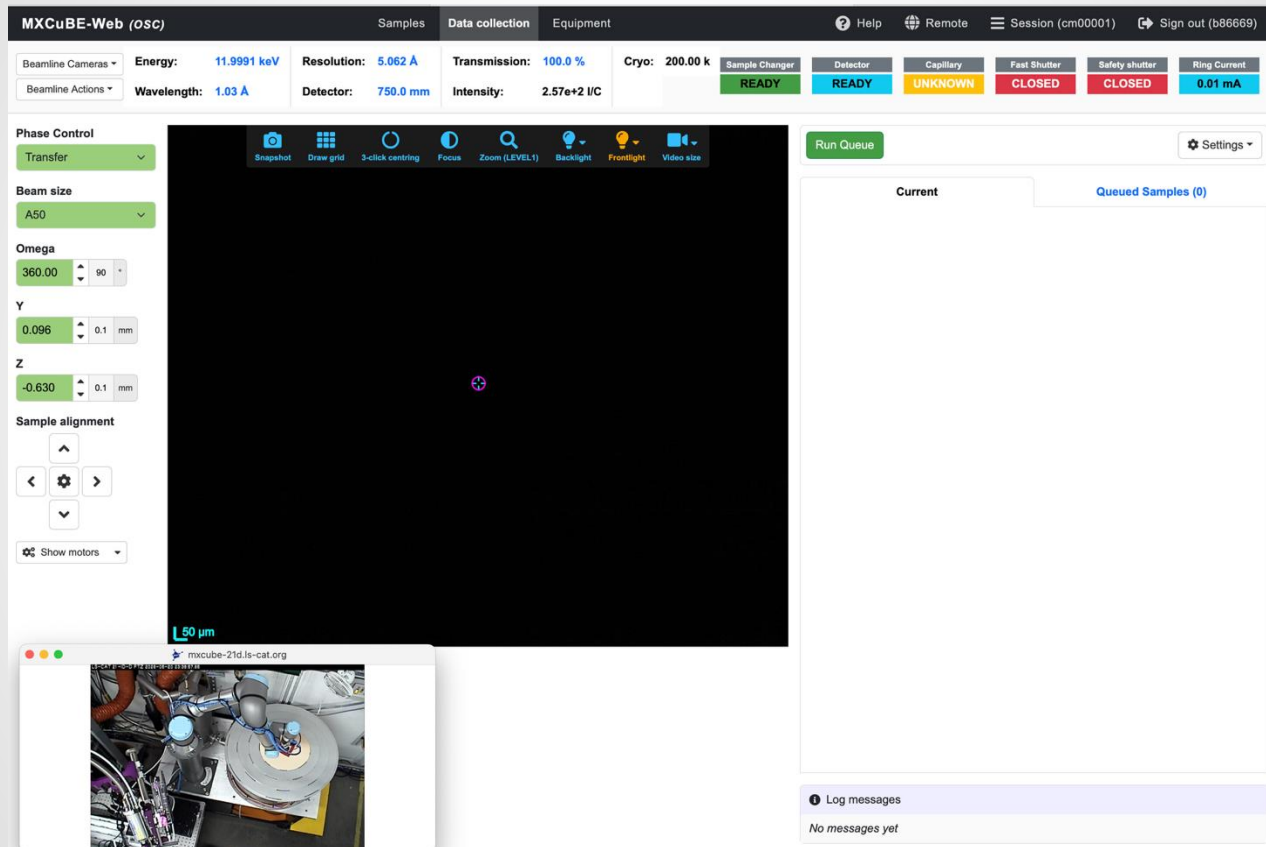
Run Queue Settings ▾

Current Queued Samples (0)

50 μm

mxcube-21d-is-cat.org

Log messages
No messages yet



MxCuBE Pre-Meeting May 21, 2026



UR7 READY



Eiger 16M READY



Status

State **IDLE** Temp: 27.7°C Humidity: 4% Buffer free: 161.1 GB

Control

Abort

Initialize

Buffer

Clear Buffer

Status message

State: IDLE | Temp: 27.7°C | Humidity: 4.0% | Buffer free: 161.1 GB

fast_shutter READY



safety_shutter READY



OPEN

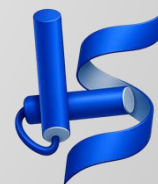
CLOSED

station_shutter READY



OPEN

CLOSED



Get samples from PyISPyB

+ Create new sample

Clear sample list

Table View

Filter :

Filter options

Add Task to Samples

Settings

Collect 0/6

Cell 6

Code : LSCAT-PUCK-001 Puck 1

DAPK - DAPK-001

6:01

DAPK - DAPK-002

6:02

AncMT - AncMT-001

6:03

AncMT - AncMT-002

6:04

DAPK - TEST-005

6:05

DAPK - TEST-006

6:06



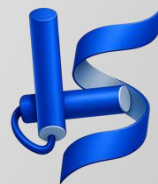
SynchWeb: Data Collections & Diffraction Viewer

The screenshot displays the SynchWeb web application interface. At the top, there is a navigation bar with links for Home, Calendar, and Logout, along with the LS-CAT logo and a status bar showing 'Logistics', 'Stats', and 'Fault Reports'. Below the navigation bar, a breadcrumb trail indicates the current location: 'Data Collections' > '21-ID-D' > 'cm00001-1'. The main content area is divided into several sections. The top section shows 'Data Collections for cm00001-1 on 21-ID-D' with a search bar and a list of tabs: 'Data Collections', 'Grid Scans', 'Full Collections', 'Auto Integrated', 'Phasing', 'Processing Errors', 'Screenings', 'Edge Scans', 'MCA Spectra', 'Robot Actions', 'Sample Actions', and 'Favourites'. The 'Data Collections' tab is active, showing a table of data collections. The table has columns for 'Sample', 'Flux', 'Q Start', 'Q Overlap', 'Detector Resolution', 'Wavelength', 'Exposure', 'Transmission', 'Beamsize', and 'Type'. The first row shows 'Sample: DAPK-002', 'Flux: 1.25e+5', 'Q Start: 0.0°', 'Q Overlap: 0.0°', 'Detector Resolution: 2.30Å', 'Wavelength: 1.0333Å (11.999eV)', 'Exposure: 0.0500s', 'Transmission: 39.43%', 'Beamsize: 0x0µm', and 'Type: OSC'. The status is 'Stopped'. Below the table, there are sections for 'Auto Processing' and 'Downstream Processing'. The 'Diffraction Image Viewer' is open, showing a diffraction pattern with concentric rings. The viewer has controls for 'Exposure', 'Transmission', 'Resolution', 'Wavelength', and 'Oscillation'. The 'View Full Screen' button is also visible. The bottom right corner of the viewer shows 'Resolution: 0.73Å' and 'Close' button.



What is next?

- Bring up 21-ID-F with MXCuBE3 (fixed-energy, side-bounce)
- Automated loop centering with Murko
- Mesh scans with Dozor scoring + x-ray centering
- Screening/characterization/collection workflows
- Energy control (currently read-only)
- SSO login authentication via Argonne National Lab



Acknowledgements

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 - Claude
 - ChatGPT
 - Gemini

